# Chapter 10: Function Operations

**MULTIPLE CHOICE**

**1.** Given the functions $f(x) = x^2 - 3$ and $g(x) = -9 - x$, determine the equation for the combined function $y = f(x) + g(x)$.

**A.** $y = x^2 - 27x - 12$
**C.** $y = x^2 + 27x + 6$
**B.** $y = x^2 - x + 6$
**D.** $y = x^2 - x - 12$

ANS: D DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: add functions | subtract functions

**2.** Given the functions $f(x) = x^2 - 8$ and $g(x) = -2 - x^2$, determine the equation for the combined function $y = f(x) + g(x)$.

**A.** $-6 - 2x^2$
**C.** $16 - x^4$
**B.** $-10$
**D.** $4 - x$

ANS: B DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: subtract functions | add functions

**3.** Given the functions $f(x) = x^3 - 81x$ and $g(x) = x + 9$, determine the simplified equation for the combined function $y = \dfrac{f(x)}{g(x)}$.

**A.** $x(x+9), x \neq -9$
**C.** $\dfrac{x}{x-9}, x \neq -9, 9$
**B.** $x(x-9), x \neq -9$
**D.** $\dfrac{x}{x+9}, x \neq -9$

ANS: B DIF: Easy OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions

**4.** Given the functions $f(x) = x^2 - 8$ and $g(x) = -5 - x$, determine an equation for the combined function $h(x) = f(g(x))$.

**A.** $y = x^2 + 10x + 17$
**C.** $y = -x^2 + 3$
**B.** $y = -x^2 - 13$
**D.** $y = x^2 - 10x + 3$

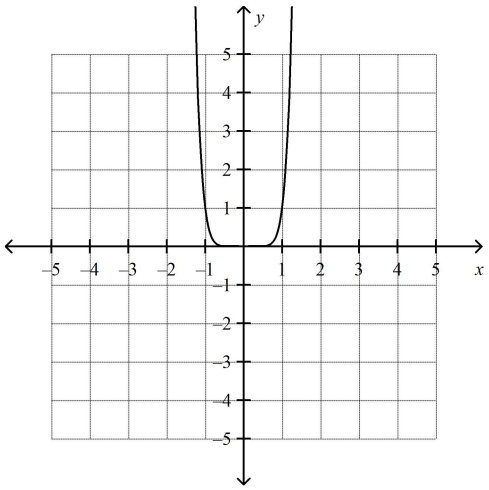
ANS: A DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: add functions | subtract functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

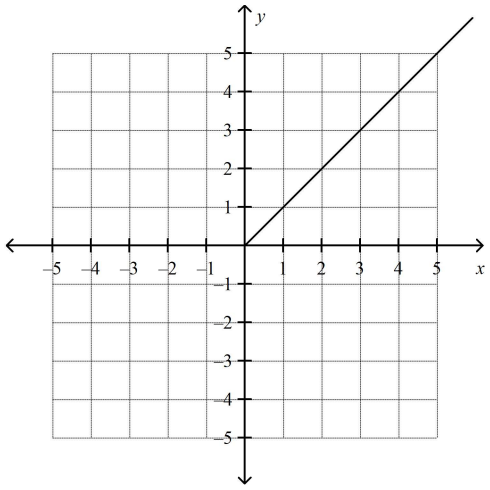
*For the following question(s), assume that x is in radians, if applicable.*

**5.** Given the functions $f(x) = x^8$ and $g(x) = \sqrt[8]{x}$, the graph of the combined function $h(x) = f(g(x))$ most likely resembles

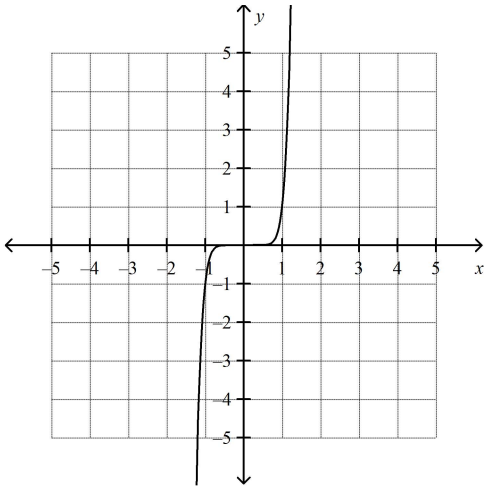
**A.**



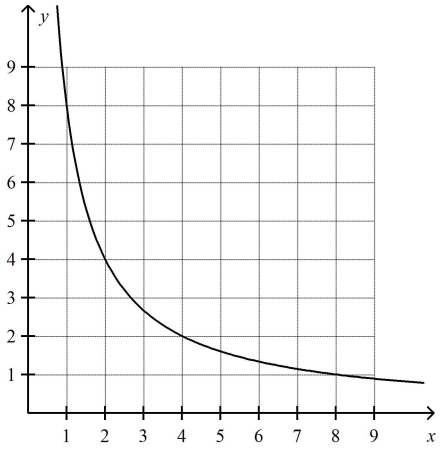

**C.**




**B.**




**D.**

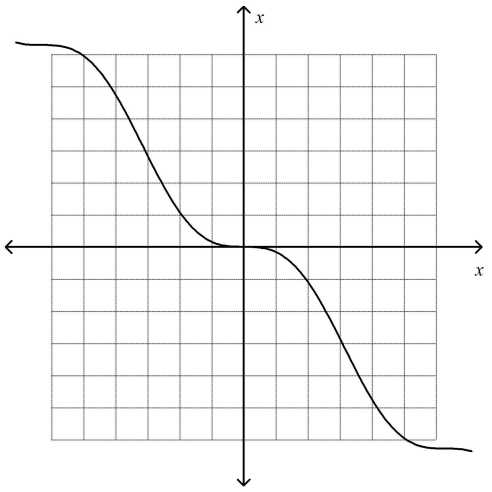



ANS: C DIF: Average OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: inverse function | composite functions

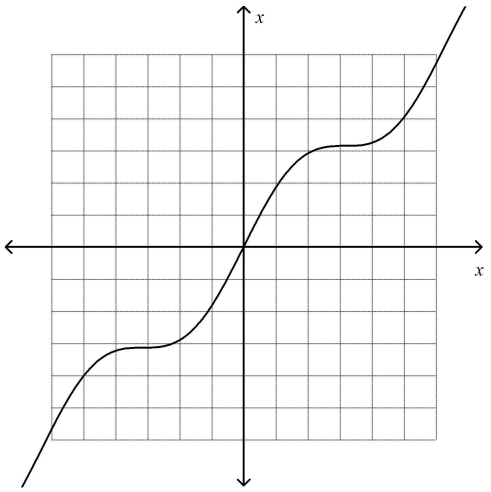
Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**6.** Given the functions $f(x) = \cos x$ and $g(x) = x$, a graph of the combined function $h(x) = f(x) - g(x)$ most likely resembles

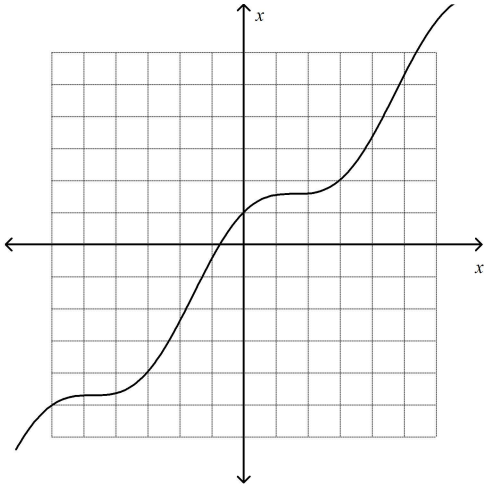
**A.**



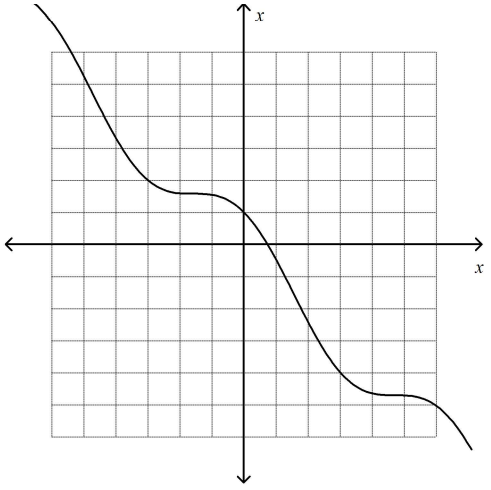

**C.**




**B.**




**D.**




ANS: D DIF: Difficult OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: add functions | graph | subtract functions

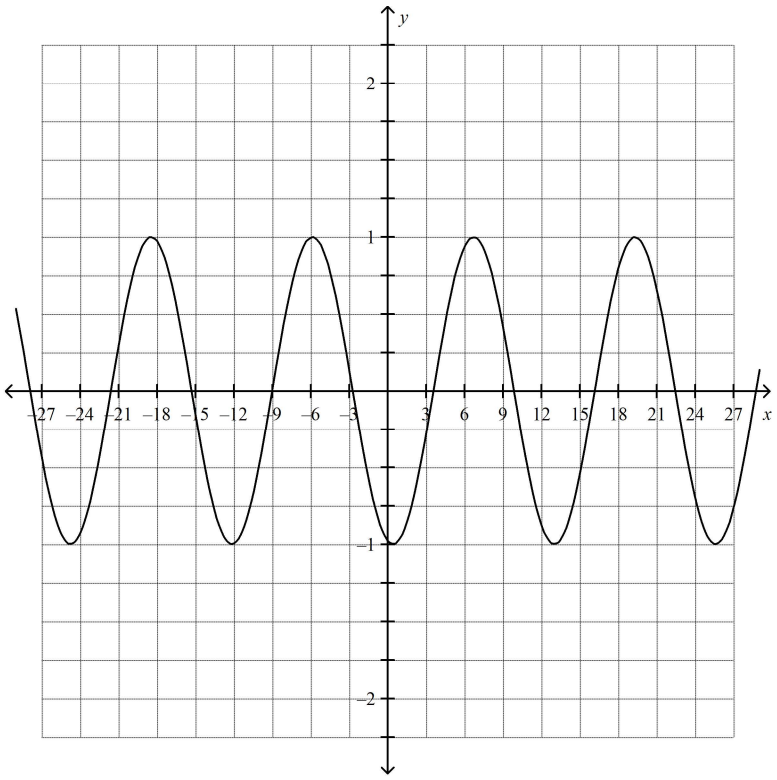
**7.** Given the functions $f(x) = 9^x$ and $g(x) = 9\sin x$, what is the range of the composite function $h(x) = f(x)g(x)$?

**A.** $\{y | -9 \le y \le 9, y \in \mathrm{R}\}$
**B.** cannot be determined
**C.** $\{y | y \in \mathrm{R}\}$
**D.** $\{y | y > 9, y \in \mathrm{R}\}$

ANS: C DIF: Easy OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: multiply functions | range

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**8.** Shown is the graph of $h(x) = f(g(x))$, where $f(x) = \sin x$ and $g(x)$ is a function of the form $g(x) = a(x + b)$.




What equation represents $g(x)$?

**A.** $g(x) = \frac{1}{2}(x + 9)$

**B.** $g(x) = 2(x - 9)$

**C.** $g(x) = 2(x + 9)$

**D.** $g(x) = \frac{1}{2}(x - 9)$

ANS: A DIF: Difficult OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: composite functions | transformations | graph

**9.** Given the functions $f(x) = \sin x$ and $g(x) = \sqrt{2 + x}$, what is the value of $g(f(\pi))$?

**A.** $\sqrt{\pi}$

**B.** $\sqrt{3}$

**C.** $\sqrt{2}$

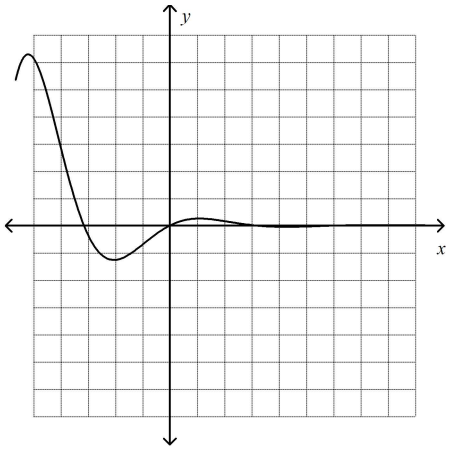
**D.** 1

ANS: C DIF: Average OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: composite functions | evaluate

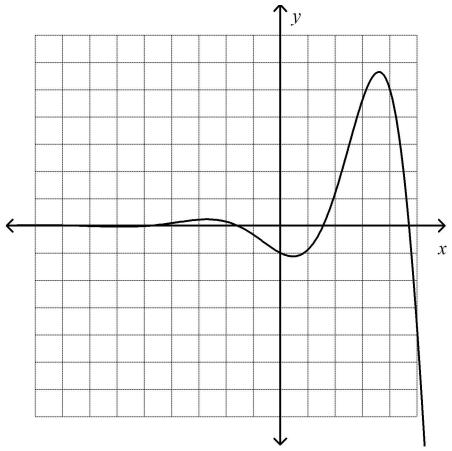
Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**10.** Given the functions $f(x) = 0.6^x$ and $g(x) = \cos x$, the graph of the combined function $h(x) = f(x)g(x)$ most likely resembles

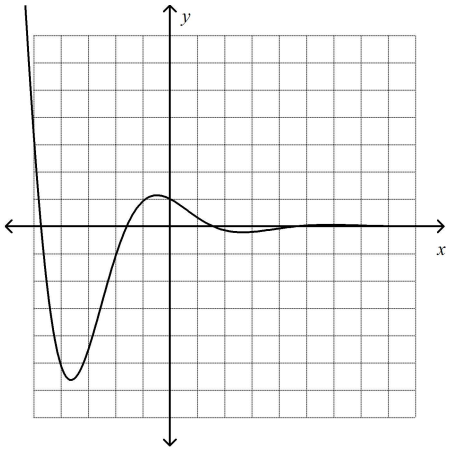
**A.**



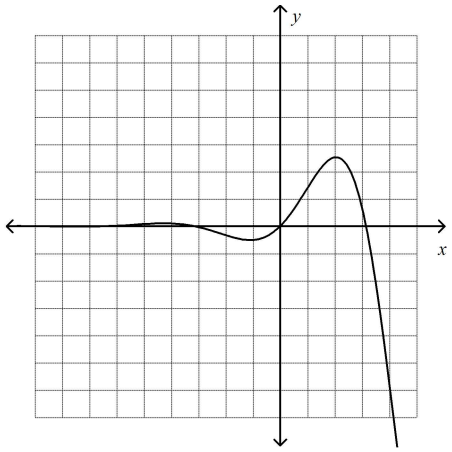

**C.**




**B.**




**D.**

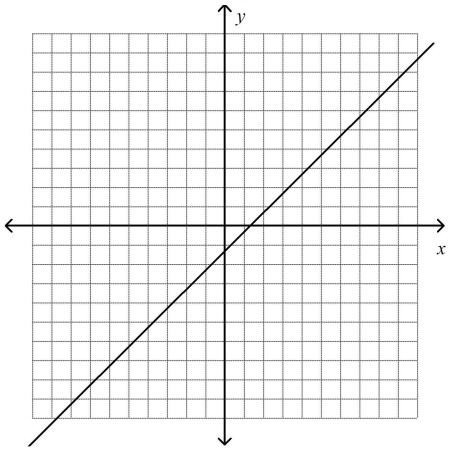



ANS: B DIF: Difficult OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: multiply functions | graph

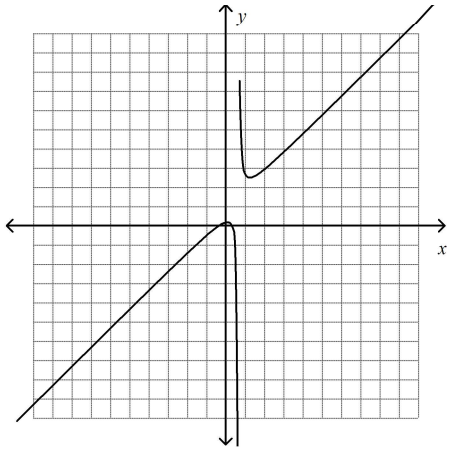
Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**11.** Given the functions $f(x) = x^2 - 4$ and $g(x) = x - 4$, a graph of the combined function $h(x) = \dfrac{f(x)}{g(x)}$ most likely resembles

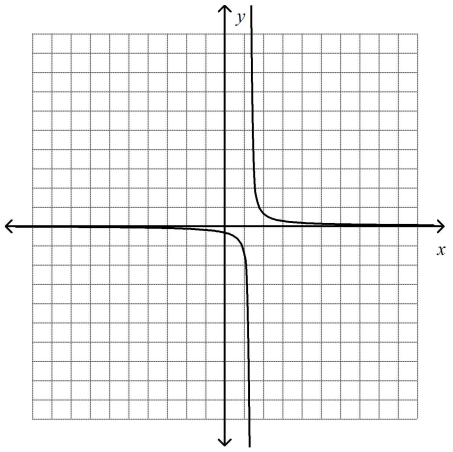
**A.**



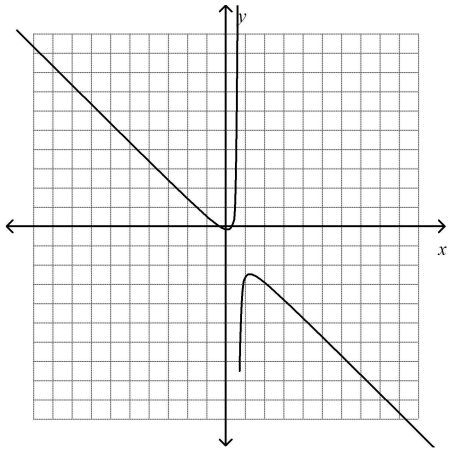

**C.**




**B.**



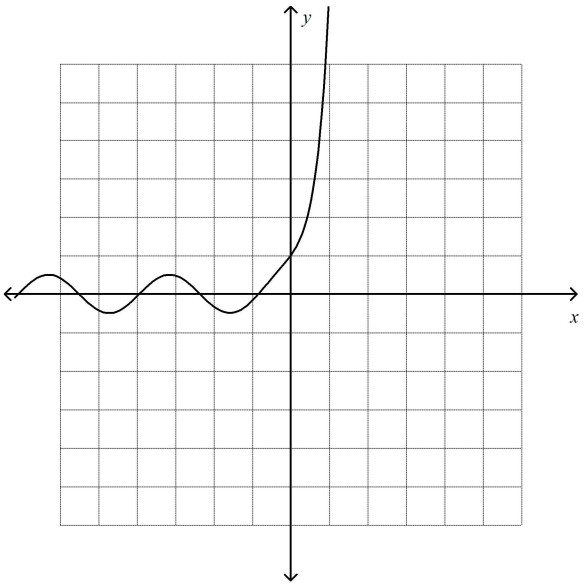

**D.**




ANS: C DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**12.** An equation for the graph shown is most likely




**A.** $4^x + \cos x$

**B.** $4^x \cos x$

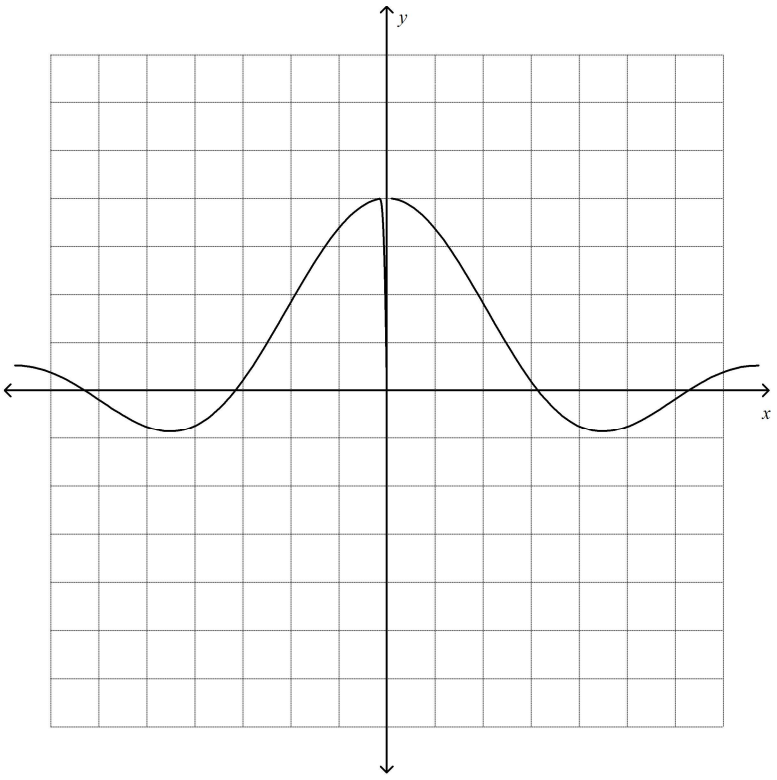
**C.** $4^x + \sin x$

**D.** $4^{\sin x}$

ANS: A DIF: Average OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: add functions | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**13.** An equation for the graph shown is most likely




**A.** $f(x) = \frac{\cos x}{x}$

**B.** $2^{\sin x}$

**C.** $\frac{\sin x}{\cos x}$

**D.** $f(x) = \frac{\sin x}{x}$

ANS: D DIF: Difficult OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions | graph

**14.** Given the functions $f(x) = -\sqrt{x+2}$ and $g(x) = x^2 - 2x + 7$, what is the domain of the combined function $k(x) = f(x) - g(x)$?

**A.** cannot be determined

**B.** $\{x \mid x \geq 7, x \in \mathrm{R}\}$

**C.** $\{x \mid x \geq -2, x \in \mathrm{R}\}$

**D.** $\{x \mid x \geq -9, x \in \mathrm{R}\}$

ANS: C DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: subtract functions | domain

**15.** Given the functions $f(x) = 8^x$ and $g(x) = \log(7 - x)$, determine the domain of the combined function $h(x) = f(x)g(x)$.

**A.** $\{x \mid x > -7, x \in \mathrm{R}\}$

**B.** cannot be determined

**C.** $\{x \mid x < 7, x \in \mathrm{R}\}$

**D.** $\{x \mid x \in \mathrm{R}\}$

ANS: C DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: multiply functions | domain

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**16.** Given the functions $f(x) = 7\cos x$ and $g(x) = x^2 - 3x - 54$, determine the domain of the combined function $h(x) = \dfrac{f(x)}{g(x)}$.

**A.** $\left\{x \mid x \neq \dfrac{\pi}{2}, x \neq \dfrac{3\pi}{2}, x \in \mathrm{R}\right\}$

**B.** $\{x \mid x \neq -9, x \neq 6, x \in \mathrm{R}\}$

**C.** $\{x \mid x \neq 1, x \neq 54, x \in \mathrm{R}\}$

**D.** $\{x \mid x \neq 9, x \neq -6, x \in \mathrm{R}\}$

ANS: D DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions | domain

**17.** Given the functions $f(x) = 5^x$ and $g(x) = 4\sin x$, what is the domain of the composite function $h(x) = f(g(x))$?

**A.** $\{x \mid -4 \leq x \leq 4, x \in \mathrm{R}\}$

**B.** $\{x \mid x > 5, x \in \mathrm{R}\}$

**C.** $\{x \mid x \in \mathrm{R}\}$

**D.** cannot be determined

ANS: C DIF: Easy OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: composite functions | domain

**18.** Given the functions $f(x) = 7\sin x$ and $g(x) = -7$, determine the range of the combined function $k(x) = f(x) + g(x)$.

**A.** $\{y \mid y \neq -7, y \in \mathrm{R}\}$

**B.** $\{y \mid y \in \mathrm{R}\}$

**C.** $\{y \mid -14 \leq y \leq 0, y \in \mathrm{R}\}$

**D.** $\{y \mid -7 \leq y \leq 7, y \in \mathrm{R}\}$

ANS: C DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: add functions | range

**19.** Given the functions $f(x) = x + 3$ and $g(x) = \dfrac{1}{x-3}$, what is the simplified form of $(f \circ g)(x)$?

**A.** $(f \circ g)(x) = \dfrac{-8}{x}, x \neq 0$

**B.** $(f \circ g)(x) = \dfrac{3x-8}{x+3}, x \neq -3$

**C.** $(f \circ g)(x) = \dfrac{3x-8}{x-3}, x \neq 3$

**D.** $(f \circ g)(x) = \dfrac{1}{x}, x \neq 0$

ANS: C DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Composite Functions KEY: composite functions | notation

**20.** If $f(x) = -7\sqrt{x} - 5$ and $g(x) = \sqrt{x} + 9$, what are the restrictions on the domain for the combined function $h(x) = \dfrac{f(x)}{g(x)}$?

**A.** $x \geq 0$

**B.** $x \geq 0, x \neq 81$

**C.** $x \neq 81$

**D.** $x \neq 3$

ANS: A DIF: Easy OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions
KEY: divide functions | domain | restrictions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**21.** If $f(x) = -3\sqrt{x} - 1$ and $g(x) = \sqrt{x} + 5$, what is the simplified combined function $h(x) = \dfrac{f(x)}{g(x)}$?

**A.** $h(x) = \dfrac{-3x + 14\sqrt{x} - 5}{x + 25}, x \geq 0$

**B.** $h(x) = \dfrac{-3x + 14\sqrt{x} + 5}{x + 25}, x \geq 0$

**C.** $h(x) = \dfrac{-3x + 14\sqrt{x} - 5}{x - 25}, x \geq 0, x \neq 25$

**D.** $h(x) = \dfrac{-3x + 14\sqrt{x} + 5}{x - 25}, x \geq 0, x \neq 25$

ANS: D DIF: Easy OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions | domain

**22.** Given $f(x) = 9x^2 + 7x$ and $g(x) = 2 - x$, determine $5g(x) - f(x)$.

**A.** $45x^2 + 36x - 2$

**B.** $9x^2 + 6x + 2$

**C.** $45x^2 + 6x - 5$

**D.** $-9x^2 - 12x + 10$

ANS: D DIF: Average OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: add functions

**23.** Given the function $f(x) = x^2 - 5$, determine the value of $f(f(-1))$.

**A.** –8

**B.** 16

**C.** 11

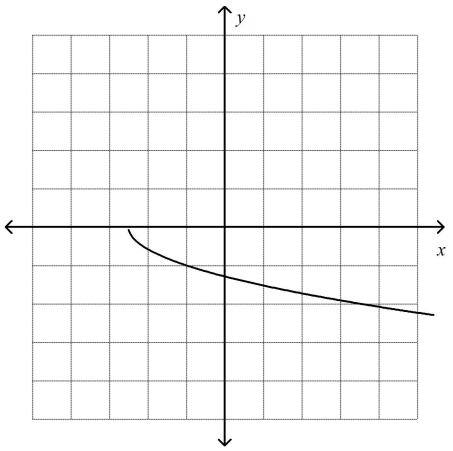
**D.** –4

ANS: C DIF: Average OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: composite functions | evaluate

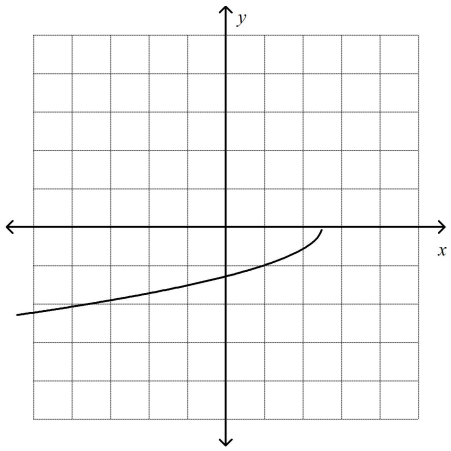
Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**24.** Given the functions $f(x) = \sqrt{x}$ and $g(x) = -\frac{1}{3}(x-5)$, which of the following is most likely the graph of $y = f(g(x))$?

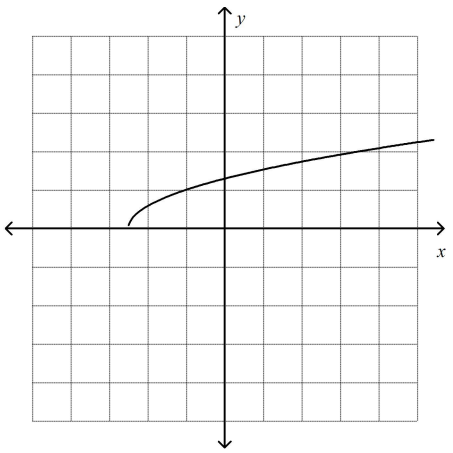
**A.**



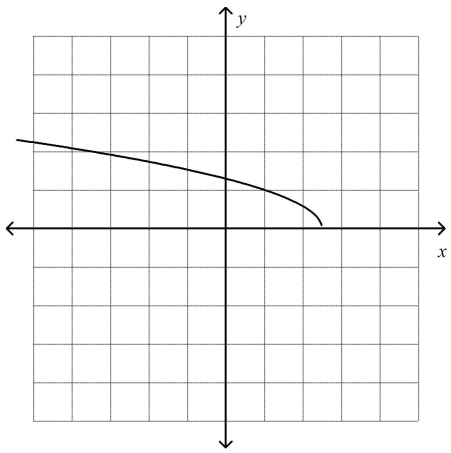

**C.**




**B.**




**D.**

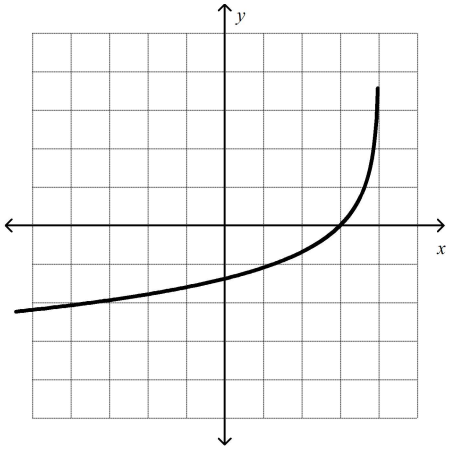



ANS: D DIF: Average OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: composite functions | transformations | graph

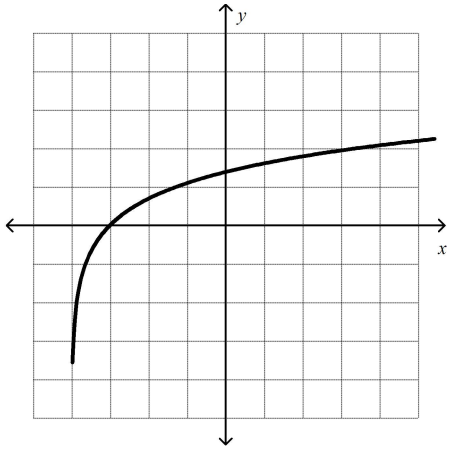
Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**25.** Given the functions $f(x) = \log x$ and $g(x) = x + 4$, which of the following is most likely the graph of $h(x) = f(g(x))$?

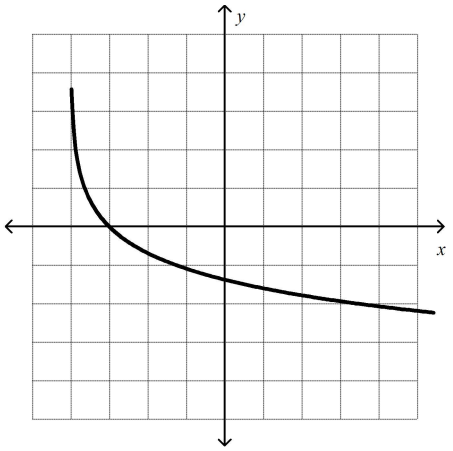
**A.**



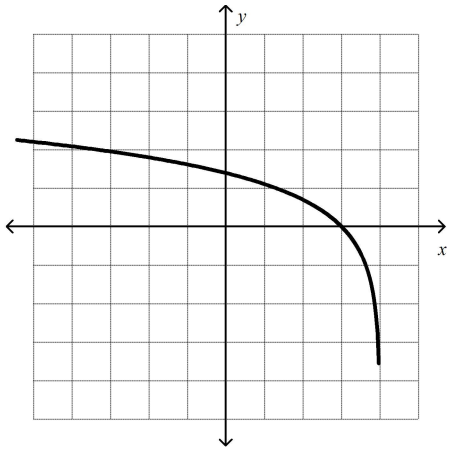

**C.**




**B.**




**D.**




ANS: C DIF: Average OBJ: Section 10.3 NAT: RF1
TOP: Composite Functions KEY: composite functions | transformations | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**MATCHING**

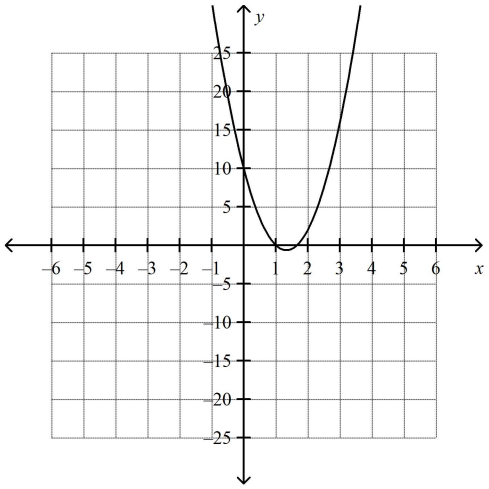
*Use the following information to answer the matching questions:*

$f(x) = (2x-2)^2$

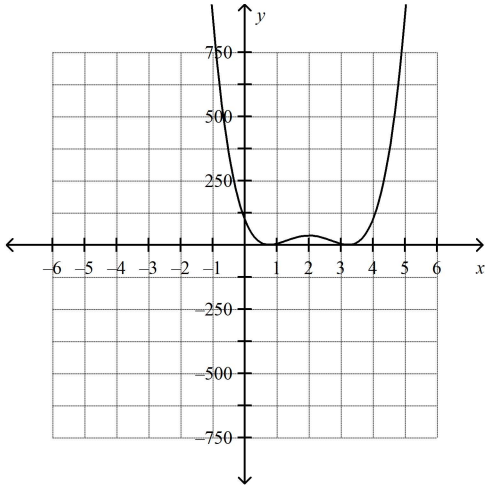
$g(x) = 2x^2 - 8x + 6$

*Match each function with its corresponding graph.*

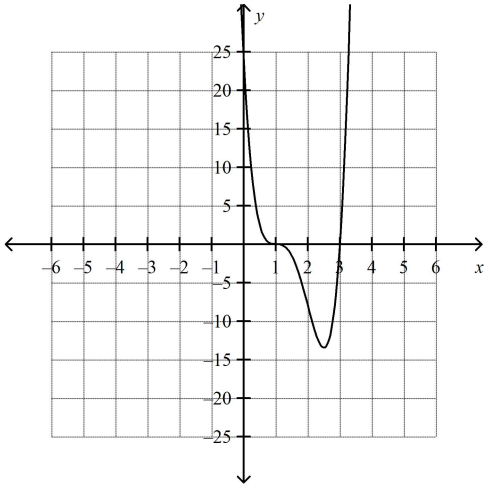
**A.**



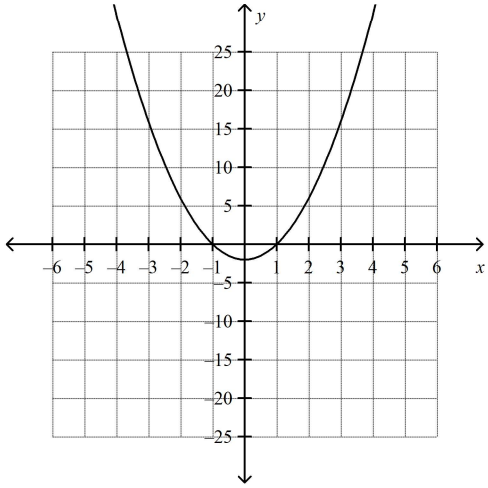

**D.**



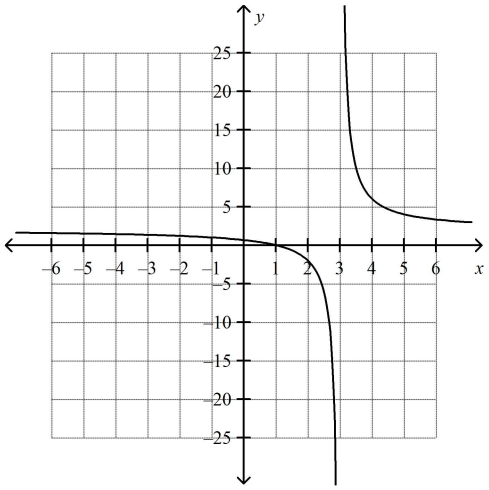

**B.**



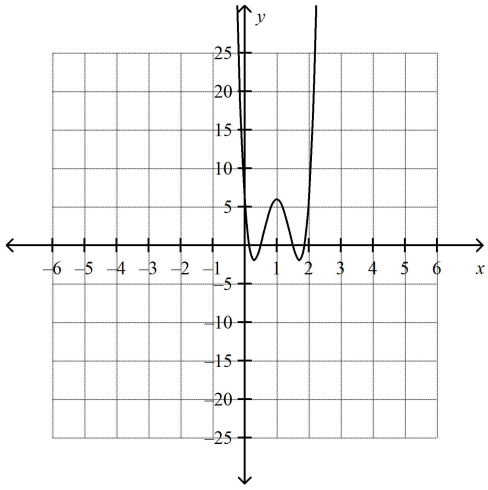

**E.**




**C.**




**F.**




**1.** $h(x) = f(x) + g(x)$

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

2. $h(x) = f(x) - g(x)$
3. $h(x) = f(x)g(x)$
4. $h(x) = \dfrac{f(x)}{g(x)}$
5. $h(x) = f(g(x))$
6. $h(x) = g(f(x))$

1. ANS: A DIF: Average OBJ: Section 10.1 NAT: RF1
   TOP: Sums and Differences of Functions KEY: add functions | graph
2. ANS: E DIF: Average OBJ: Section 10.1 NAT: RF1
   TOP: Sums and Differences of Functions KEY: subtract functions | graph
3. ANS: B DIF: Average OBJ: Section 10.2 NAT: RF1
   TOP: Products and Quotients of Functions KEY: multiply functions | graph
4. ANS: C DIF: Average OBJ: Section 10.2 NAT: RF1
   TOP: Products and Quotients of Functions KEY: divide functions | graph
5. ANS: D DIF: Average OBJ: Section 10.3 NAT: RF1
   TOP: Composite Functions KEY: composite functions | graph
6. ANS: F DIF: Average OBJ: Section 10.3 NAT: RF1
   TOP: Composite Functions KEY: composite functions | graph

**COMPLETION**

1. If $f(x)$ is a quadratic function with positive coefficients and $g(x)$ is a linear function with positive coefficients, then $f(x) + g(x)$ is a _____ function.

   ANS: quadratic

   DIF: Easy OBJ: Section 10.1 NAT: RF1
   TOP: Sums and Differences of Functions KEY: add functions

2. If $f(x)$ is a linear function and $g(x)$ is a linear function, then $f(x)g(x)$ is a _____ function.

   ANS: quadratic

   DIF: Average OBJ: Section 10.2 NAT: RF1
   TOP: Products and Quotients of Functions KEY: multiply functions

3. If $f(x) = \sqrt{x-4}$ and $g(x) = x^2 - 3$, then the domain of the function $h(x) = f(g(x))$ is _____.

   ANS: $\{x \mid x \le -\sqrt{7} \text{ or } x \ge \sqrt{7}, x \in \mathrm{R}\}$

   DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
   KEY: composite functions | domain

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**4.** If $f(x) = \sqrt{x-8}$ and $g(x) = x^2 + 4$, then the domain of the function $h(x) = g(f(x))$ is _____ .

ANS: $\{x | x \geq 8, x \in \mathrm{R}\}$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | domain

**5.** If $h(x) = f(g(x))$, and $h(x) = 7\cot^2 x$, then $f(x)$ = _____ and $g(x)$ = _____.

ANS:
Answers may vary. Sample answer:
$\frac{7}{x^2}$; $\tan x$

DIF: Difficult OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions

**SHORT ANSWER**

**1.** Given the functions $f(x) = x + 1$ and $g(x) = x^2 + 3x + 1$, determine a simplified equation for $h(x) = f(x) + g(x)$.

ANS:
$h(x) = f(x) + g(x)$

$= x + 1 + x^2 + 3x + 1$

$= x^2 + 4x + 2$

DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: add functions

**2.** Given the functions $f(x) = x^2 + 3$ and $g(x) = x^3 - x$, determine a simplified equation for $h(x) = f(x) - g(x)$.

ANS:
$h(x) = f(x) - g(x)$

$= x^2 + 3 - (x^3 - x)$

$= -x^3 + x^2 + x + 3$

DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: subtract functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**3.** Given the functions $f(x) = x^2 - 7$ and $g(x) = 2 - x^3$, determine a simplified equation for $h(x) = f(x)g(x)$.

ANS:

$$\begin{aligned} h(x) &= f(x)g(x) \\ &= (x^2 - 7)(2 - x^3) \\ &= -x^5 + 7x^3 + 2x^2 - 14 \end{aligned}$$

DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: multiply functions

**4.** Given the functions $f(x) = x^2 - 4$ and $g(x) = x^2 - 3x + 2$, determine a simplified equation for $h(x) = \dfrac{f(x)}{g(x)}$.

ANS:

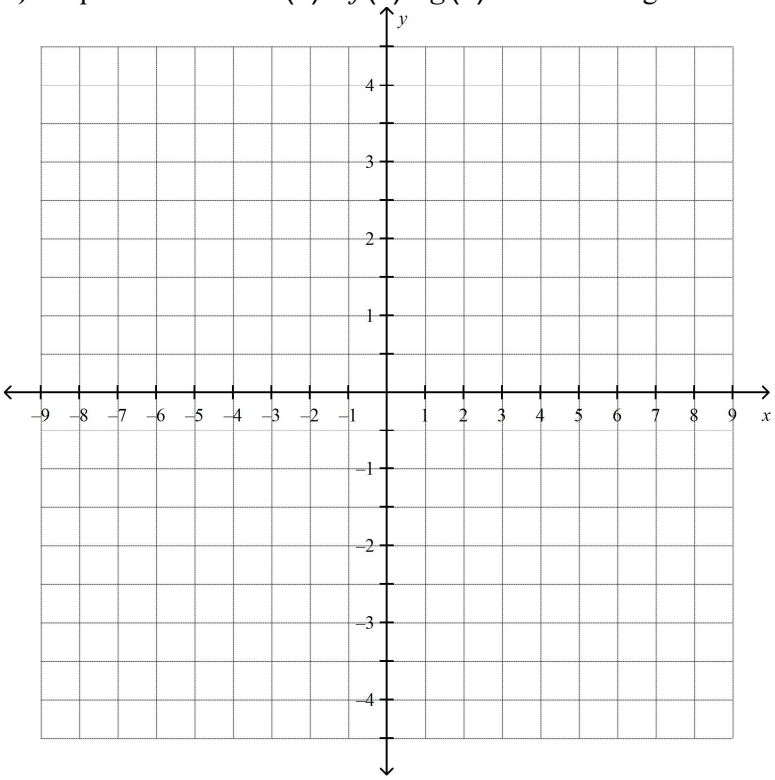
$$\begin{aligned} h(x) &= \frac{f(x)}{g(x)} \\ &= \frac{x^2 - 4}{x^2 - 3x + 2} \\ &= \frac{(x-2)(x+2)}{(x-2)(x-1)} \\ &= \frac{(x+2)}{(x-1)}, x \neq 2, x \neq 1 \end{aligned}$$

DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions | restrictions

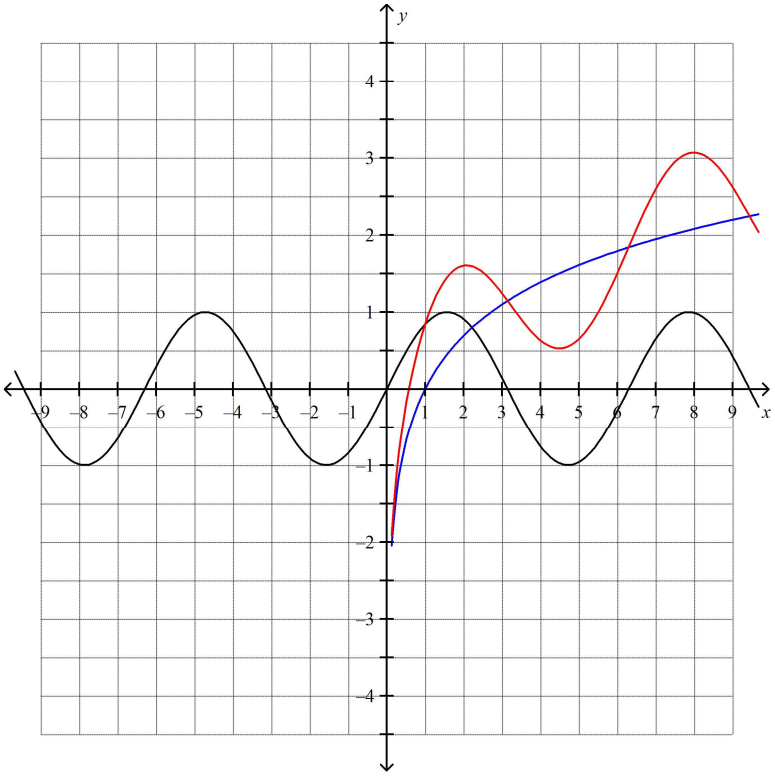
Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

*For the following question(s), assume that x is in radians, if applicable.*

**5.** **a)** Graph the functions $f(x) = \sin(x)$ and $g(x) = \log(x)$ on the grid.
**b)** Graph the function $h(x) = f(x) + g(x)$ on the same grid.




ANS:




DIF: Difficult OBJ: Section 10.1 NAT: RF1

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

TOP: Sums and Differences of Functions KEY: add functions | graph

**6.** Determine the domain and range of the function $y = \frac{\sin x}{x^2 - x - 6}$.

ANS:
Domain: $\{x | x \neq -2, x \neq 3, x \in \mathrm{R}\}$
Range: $\{y \mid y \in \mathrm{R}\}$

DIF: Average OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: divide functions | domain

**7.** Given the functions $f(x) = \cos x$ and $g(x) = 2^x + 5$, determine the value of $g(f(\pi))$.

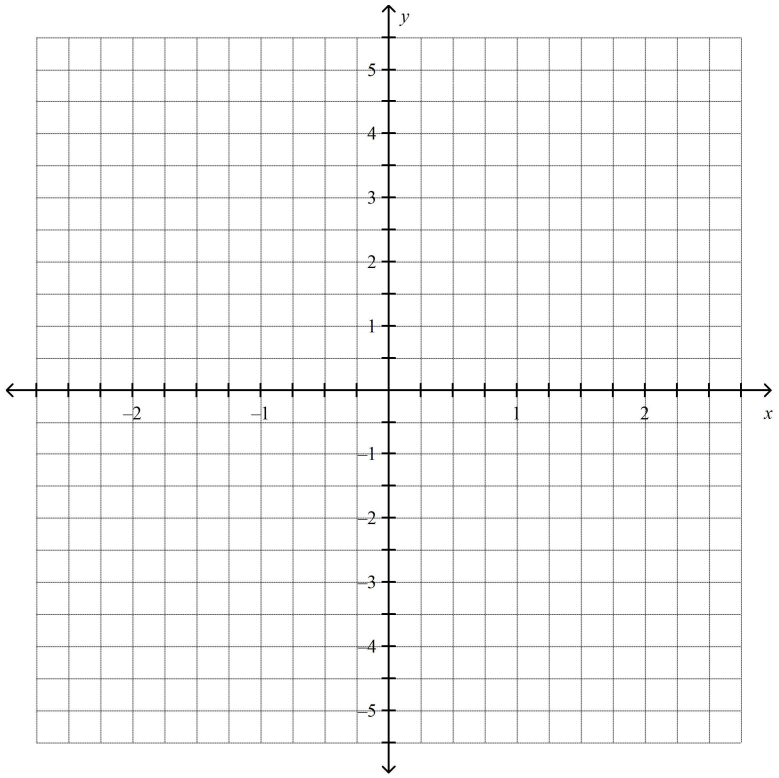
ANS:

$$g(f(x)) = g(\cos x)$$
$$= 2^{\cos x} + 5$$
$$g(f(\pi)) = 2^{(\cos \pi)} + 5$$
$$= 2^{-1} + 5$$
$$= \frac{1}{2} + 5$$
$$= \frac{11}{2} \text{ or } 5.5$$

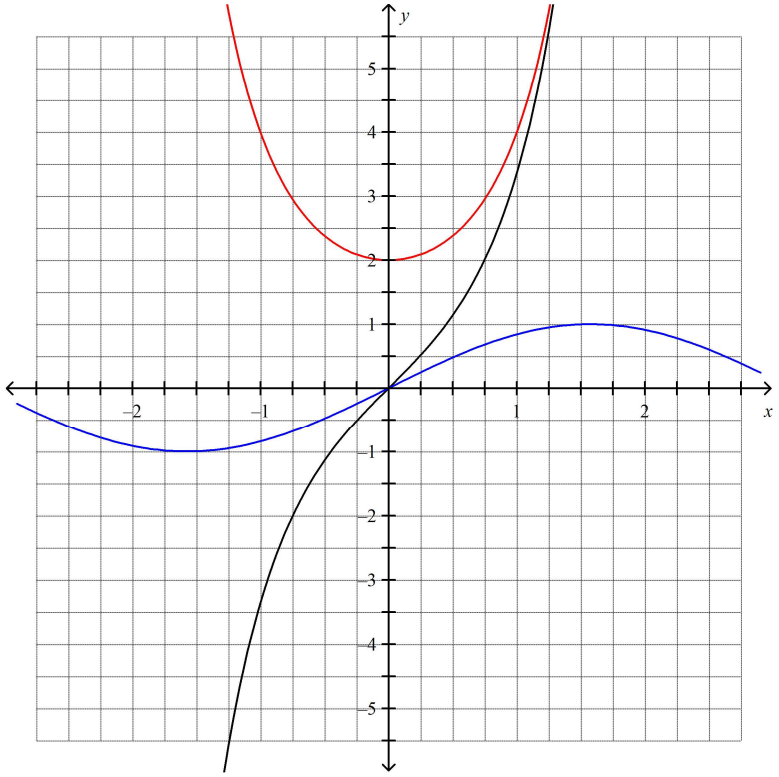
DIF: Difficult OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | evaluate

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**8.** **a)** Graph the functions $f(x) = \sin(x)$ and $g(x) = 2^{\left(x^2+1\right)}$.
**b)** Use the graphs to graph the function $h(x) = f(x)g(x)$.




ANS:




$f(x)$ is in blue, $g(x)$ is in red, and $h(x)$ is in black.

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

DIF: Difficult OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions KEY: multiply functions | graph

**9.** Determine the equation(s) of the vertical asymptote(s) of the function $y = \frac{0.9^x}{2x^2 - 3}$.

ANS:

$$2x^2 - 3 = 0$$

$$2x^2 = 3$$

$$x^2 = \frac{3}{2}$$

$$x = \pm\sqrt{\frac{3}{2}}$$

$$x = \pm\frac{\sqrt{6}}{2}$$

DIF: Difficult + OBJ: Section 10.2 NAT: RF1
TOP: Products and Quotients of Functions
KEY: divide functions | vertical asymptotes

**10.** Given the functions $f(x) = x^2 - 7$ and $g(x) = 2 - x^3$, what is the value of $f(g(2))$?

ANS:

$$g(2) = 2 - (2)^3$$

$$= -6$$

$$f(-6) = (-6)^2 - 7$$

$$= 29$$

$$f(g(2)) = 29$$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | evaluate

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**PROBLEM**

**1.** Given the functions $f(x) = x^2 + 3x + 2$ and $g(x) = x + 1$, determine a simplified equation for $k(x) = f(g(x))$.

ANS:

$$\begin{aligned} k(x) &= (x+1)^2 + 3(x+1) + 2 \\ &= x^2 + 2x + 1 + 3x + 3 + 2 \\ &= x^2 + 5x + 6 \end{aligned}$$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | evaluate

**2.** Given the functions $f(x) = \dfrac{1}{1-x^2}$ and $g(x) = \sin x$, determine the equation for $h(x) = f(g(x))$.

ANS:

$$\begin{aligned} h(x) &= \frac{1}{1-\sin^2 x} \\ &= \frac{1}{\cos^2 x} \\ &= \sec^2 x \end{aligned}$$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions

**3.** Given the functions $f(x) = x + 2$ and $g(x) = x^2$, what values of $x$ result in $f(g(x)) = g(f(x))$?

ANS:

$$\begin{aligned} f(g(x)) &= g(f(x)) \\ f(x^2) &= g(x+2) \\ x^2 + 2 &= (x+2)^2 \\ x^2 + 2 &= x^2 + 4x + 4 \\ -2 &= 4x \\ x &= -\frac{1}{2} \end{aligned}$$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**4.** Given the functions $f(x) = \sqrt{x}$ and $g(x) = x + 1$, determine all possible values of $x$ for which $f(g(x)) = g(f(x))$.

ANS:

$$f(g(x)) = g(f(x))$$

$$f(x+1) = g\left(\sqrt{x}\right)$$

$$\sqrt{x+1} = \sqrt{x} + 1$$

$$\left(\sqrt{x+1}\right)^2 = \left(\sqrt{x} + 1\right)^2$$

$$x + 1 = x + 2\sqrt{x} + 1$$

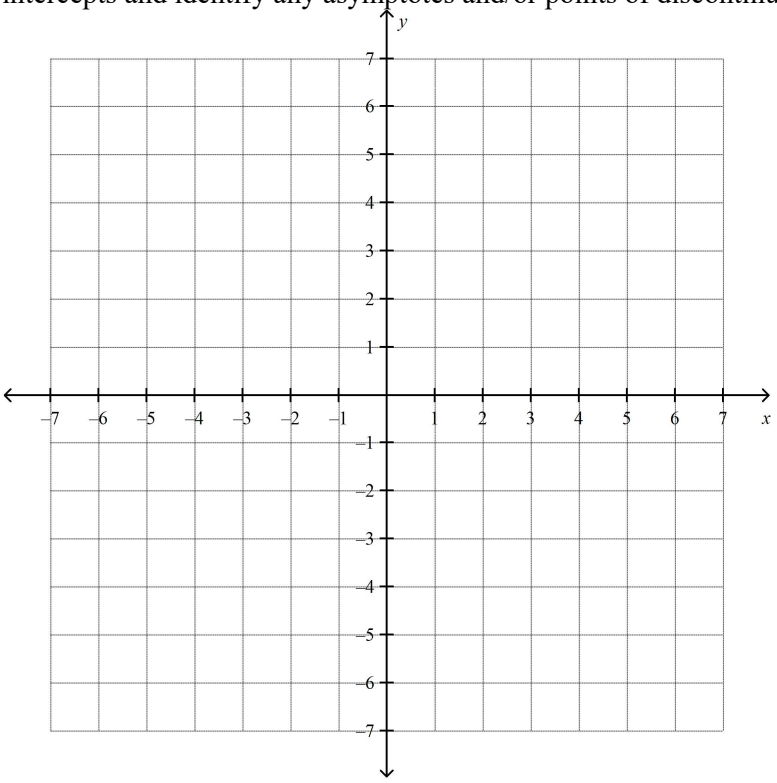
$$0 = 2\sqrt{x}$$

$$x = 0$$

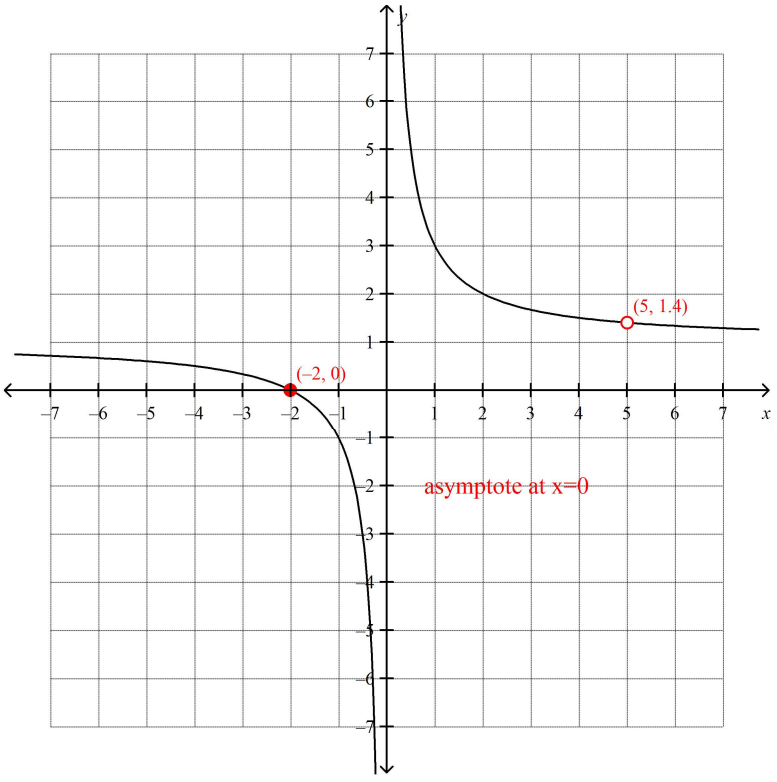
DIF: Difficult OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**5.** Given the functions $f(x) = x^2 - 3x - 10$ and $g(x) = x^2 - 5x$, graph the function $h(x) = \dfrac{f(x)}{g(x)}$. Label all intercepts and identify any asymptotes and/or points of discontinuity.




ANS:




DIF: Difficult OBJ: Section 10.2 NAT: RF1

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

TOP: Products and Quotients of Functions
KEY: divide functions | vertical asymptotes | hole

**6.** What are the domain and range of the function $y = 2^{\sin x}$, where $x$ is in radians?

ANS:
Domain: $\{x | x \in \mathrm{R}\}$
Range: $\left\{y \mid \frac{1}{2} \le y \le 2, y \in \mathrm{R}\right\}$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | domain | range

**7.** Given that $f(x) = \sqrt{x}$ and $g(x) = x^2 + 10x + 24$, what are the domain and range of the function $f(g(x))$?

ANS:
Domain: $\{x | x \le -6, x \ge -4, x \in \mathrm{R}\}$
Range: $\{y | y \ge 0, y \in \mathrm{R}\}$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | domain | range

**8.** Determine the equation for the composite function $y = f(f(x))$ given that $f(x) = \frac{1}{x}$.

ANS:
$y = f(f(x))$

$= f\left(\frac{1}{x}\right)$

$= \frac{1}{\frac{1}{x}}$

$= x, x \ne 0$

DIF: Easy OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**9.** The revenue function for a school group selling $n$ bookmarks is given by $R(n) = 2n$, and the total cost function is given by $C(n) = 144 + 0.80n$. Determine the number of bookmarks that need to be sold for the school group to break even.

ANS:
The profit function is given by $P(n) = R(n) - C(n)$.

$$P(n) = 2n - (144 + 0.80n)$$
$$= 1.2n - 144$$

To break even, $P(n) = 0$.

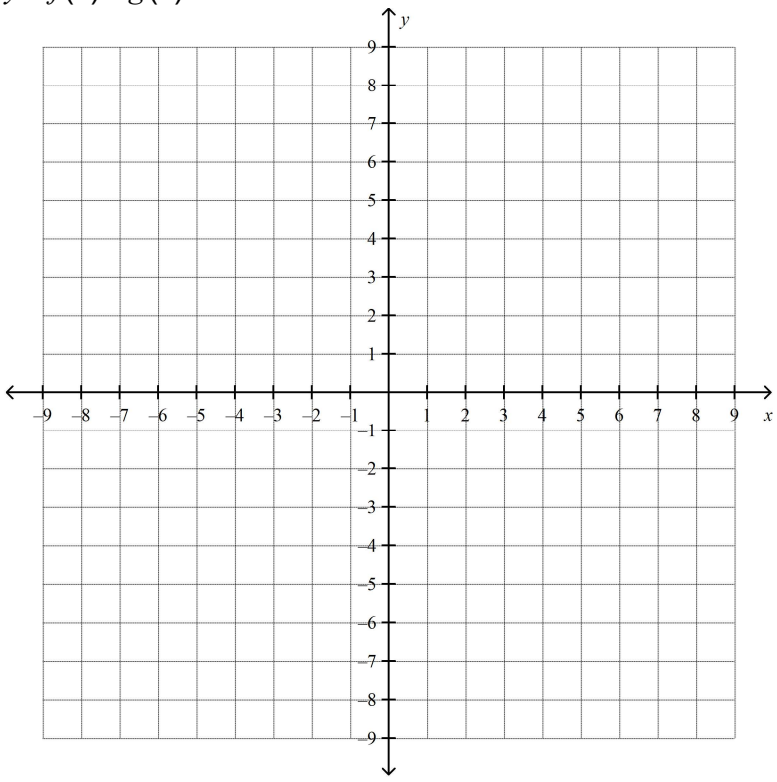
$$1.2n - 144 = 0$$
$$1.2n = 144$$
$$n = 120$$

The number of bookmarks that need to be sold to break even is 120.

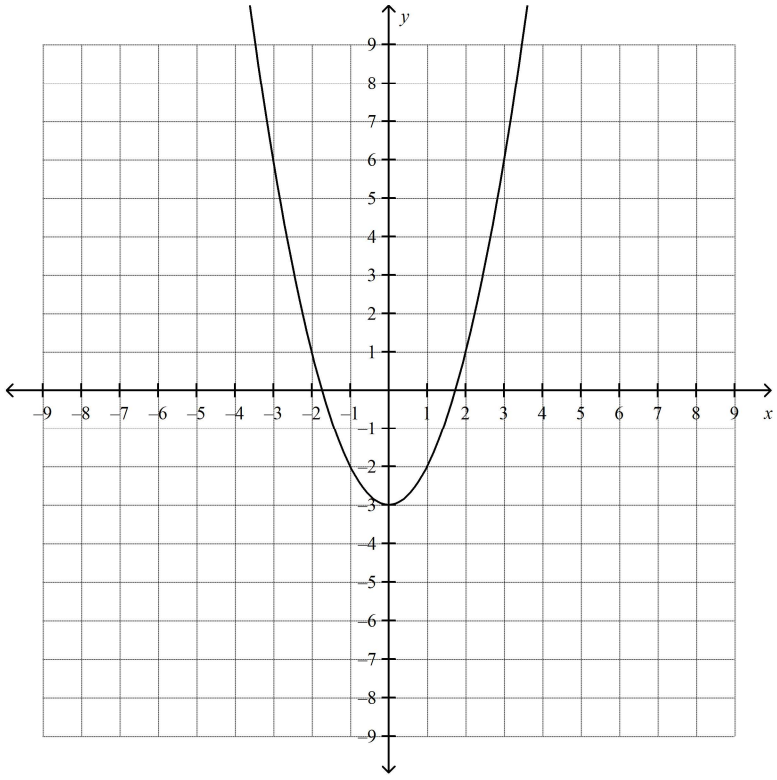
DIF: Average OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: subtract functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**10.** Given the functions $f(x) = x^2$ and $g(x) = 3$, sketch the graph of the combined function $y = f(x) - g(x)$.




ANS:



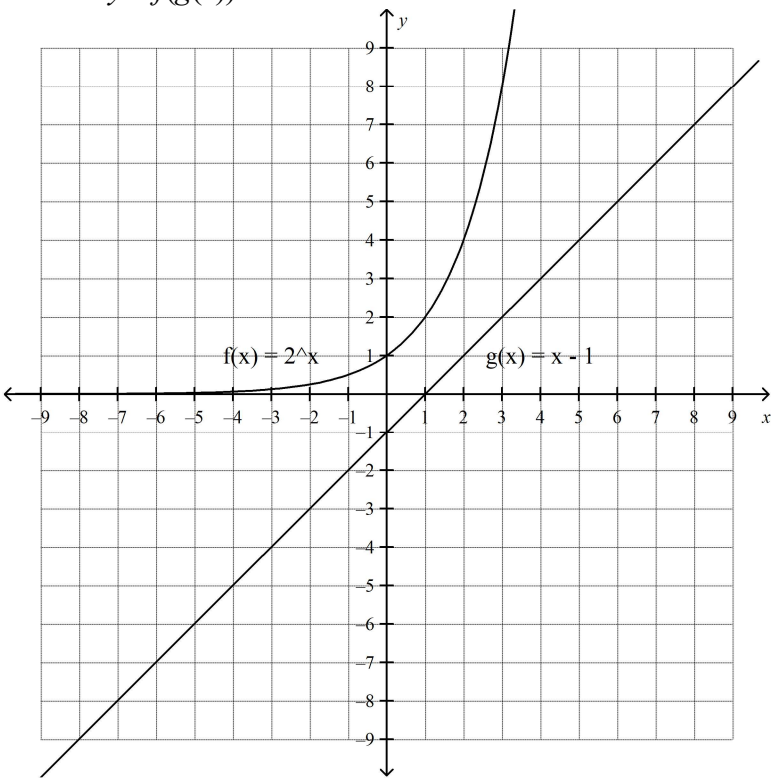

DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

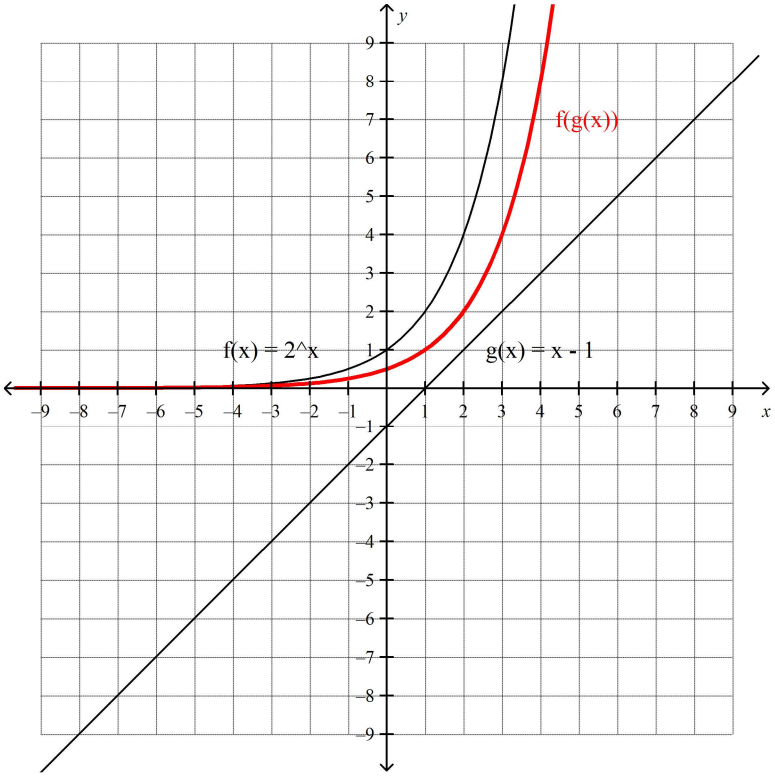
KEY:  combined function | difference | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**11.** Given the graphs of the functions $f(x) = 2^x$ and $g(x) = x - 1$, sketch the graph of the combined function $y = f(g(x))$.




ANS:




DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

KEY: composite functions | graph | transformations

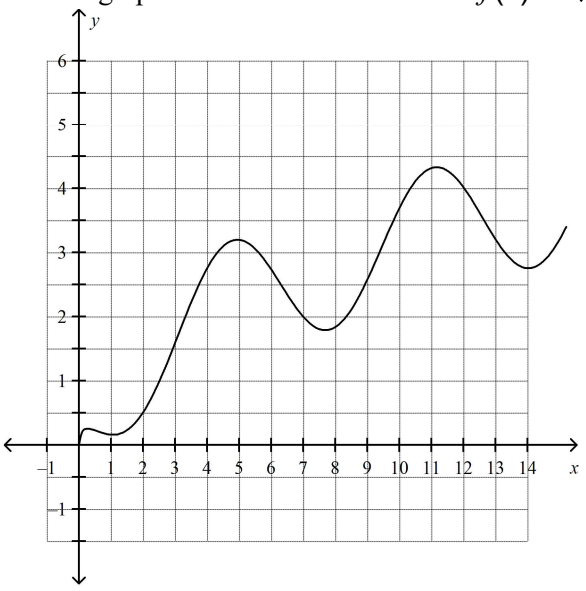
**12.** The revenue function for a company selling $n$ key chains is given by $R(n) = 2n$, and the total cost function is given by $C(n) = 100 + 0.80n$. Determine the simplified equation for $P(n)$, the profit function for the company.

ANS:

$$\begin{aligned} P(n) &= R(n) - C(n) \\ &= 2n - (100 + 0.80n) \\ &= 1.2n - 100 \end{aligned}$$

DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: profit function | subtract functions

**13.** Use the graph of the combined function $f(x) = \sqrt{x} - \sin x$ to solve the inequality $\sin x > \sqrt{x}$.

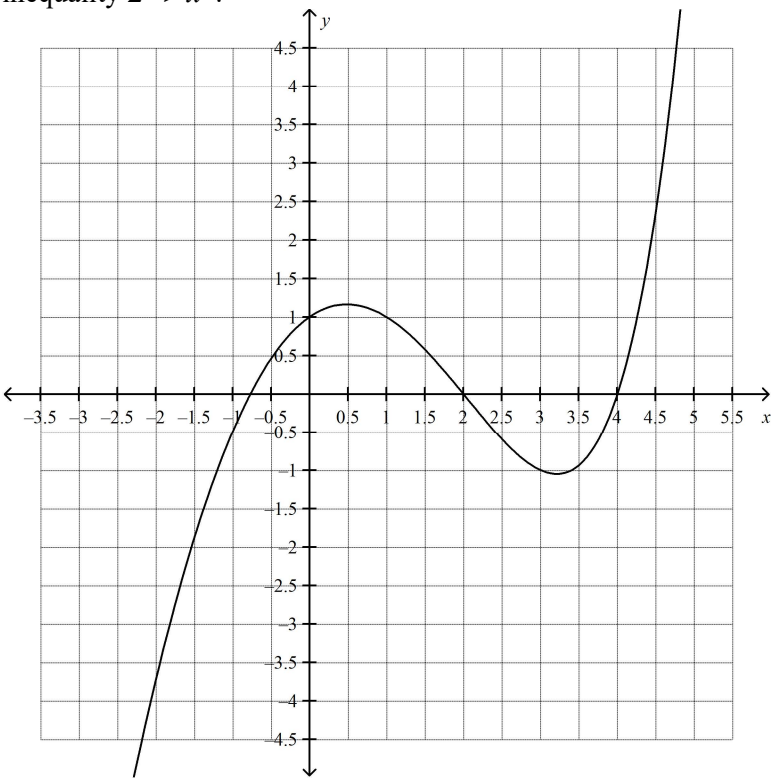



ANS:
The given inequality is never satisfied.
There is no solution.

DIF: Difficult + OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: combined function | inequality | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**14.** Use the graph of the combined function $f(x) = 2^x - x^2$ to determine an approximate solution to the inequality $2^x > x^2$.




ANS:
From the graph, the *x*-values that result in the combined function having a value greater than zero (positive), which is equivalent to $2^x > x^2$, are $-0.8 < x < 2$ or $x > 4$.

DIF: Difficult + OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: combined function | inequality | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**15.** Jenny and Jimmy are a married couple who work at the same store. Jimmy's total weekly salary, in dollars, if he sells $x$ items is given by $S(x) = 10.0 + 5x$, and Jenny's total weekly salary, in dollars, if she sells $x$ items is given by $S(x) = 8.0 + 6x$.
**a)** Assuming that they sell the same number of items in a week, what is the minimum number of items they have to sell so that Jenny's weekly salary is at least \$100 more than Jimmy's?
**b)** Assuming that they sell the same number of items in a week, what is the minimum number of items they each need to sell to make their combined weekly salary greater than \$1000?

ANS:
**a)** $D(x) = (8.0 + 6x) - (10.0 + 5x)$

$= -2.0 + x$

This difference must be at least 100.

$-2 + x \geq 100$

$x \geq 102$

The minimum number sold is 102 items.
**b)** $T(x) = (8.0 + 6x) + (10.0 + 5x)$

$= 18.0 + 11x$

This total must be at least 1000.

$18 + 11x \geq 1000$

$11x \geq 982$

$x \geq 89.272\ldots$

The minimum number sold is 90 items.

DIF: Difficult OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions
KEY: add functions | subtract functions

**16.** The revenue function for a company selling $n$ coffee mugs can be modelled by the function $R(n) = 10n$, and the total cost function can be modelled by the function $C(n) = 360 + 6n$.
**a)** Determine a simplified equation for $P(n)$, the profit function for the company.
**b)** How many coffee mugs must be sold to make a profit of \$1000?

ANS:
**a)** $P(n) = R(n) - C(n)$

$= 10n - (360 + 6n)$

$= 4n - 360$

**b)** $P(n) = 4n - 360$

$1000 = 4n - 360$

$1360 = 4n$

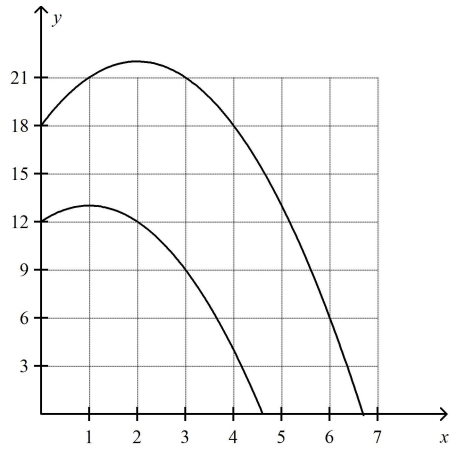
$n = 340$

340 mugs must be sold to make a profit of \$1000.

DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: subtract functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**17.** The heights, $h$, of two balls, in metres, for a horizontal distance of $x$ metres are shown in the graph.



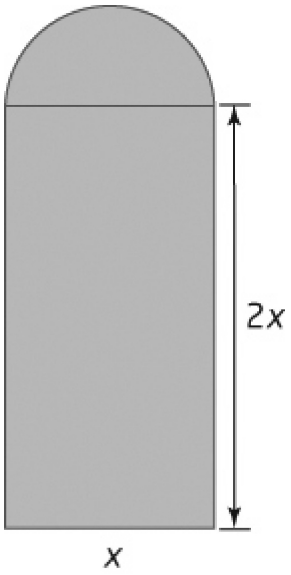

What was the difference in height of the two balls when the horizontal distance was 0 m?

ANS:
The difference in the $y$-intercepts is $18 - 12 = 6$.
One ball was 6 m above the other.

DIF: Easy OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: subtract functions | graph

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**18.** The dimensions of a window are shown.




**a)** What function in simplest form represents the area of the entire window?
**b)** If the width, $x$, of the window is 1.2 m, what is the total area of the window, to the nearest tenth of a square metre?

ANS:
**a)** Area of rectangle = $2x^2$

Area of semi-circle = $\dfrac{\pi\left(\frac{x}{2}\right)^2}{2}$

$= \dfrac{\pi x^2}{8}$

Total area = $2x^2 + \dfrac{\pi x^2}{8}$

$= \dfrac{16x^2 + \pi x^2}{8}$

$= \dfrac{x^2(16+\pi)}{8}$

**b)** Area = $\dfrac{x^2(16+\pi)}{8}$

$= \dfrac{1.2^2(16+\pi)}{8}$

$\approx 3.4$

The area of the window is approximately 3.4 m$^2$.

DIF: Average OBJ: Section 10.1 NAT: RF1
TOP: Sums and Differences of Functions KEY: add functions

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.

**19.** A vehicle depreciates 20% in value in the first year and 10% in each year after the first.
**a)** Write a function, $v_1(c)$, to describe the value of the vehicle after 1 year in terms of the original cost, $c$.
**b)** Write a function, $v_n(c)$, to describe the value after $n$ years in terms of the original cost, $c$.
**c)** If the original value of the vehicle is \$45 000, what is its value after 5 years?

ANS:
**a)** $v_1(c) = 0.8c$

**b)** $v_n(c) = (0.8c)(0.9)^n$, $n$ = number of years after the first year

**c)** $v_n(c) = (0.8(45\,000))(0.9)^4$

$= 36\,000(0.9)^4$

$= 23\,619.6$

After 5 years the vehicle will have a value of \$23 619.60.

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | depreciation

**20.** **a)** Write a function to represent the area of a circle in terms of the radius.
**b)** Write a function to represent the radius of a circle in terms of the circumference.
**c)** What is the function that represents the area of a circle in terms of the circumference?

ANS:
**a)** $A(r) = \pi r^2$

**b)** $r(C) = \dfrac{C}{2\pi}$

**c)** $A(r(C)) = \pi\left(\dfrac{C}{2\pi}\right)^2$

$= \dfrac{\pi C^2}{4\pi^2}$

$= \dfrac{C^2}{4\pi}$

DIF: Average OBJ: Section 10.3 NAT: RF1 TOP: Composite Functions
KEY: composite functions | area of circle

Copyright 2012 McGraw-Hill Ryerson. All rights reserved.